

Bonus Question - Solution

- Q (i) A vector space is a set V with a map
$$+' : V \times V \rightarrow V$$
$$(v_1, v_2) \rightarrow (v_1 + v_2)$$

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- (i) A vector space is a set V , with the map

(a)
$$+' : V \times V \rightarrow V$$

defined as $(v_1, v_2) \rightarrow (v_1 + v_2) \quad \forall v_1, v_2 \in V$

This is called vector addition

(b)
$$'\cdot' : F \times V \rightarrow V$$

defined as $(\alpha, v) \rightarrow \alpha \cdot v \quad \forall \alpha \in F$ (a scalar from field F) and $v \in V$

— (10.5)

This is called scalar multiplication.

V is called F -vector space or vector space over field F if the following properties are satisfied:

(i) $(V, +)$ is an Abelian group

(ii) $\alpha \cdot (v_1 + v_2) = \alpha \cdot v_1 + \alpha \cdot v_2 \quad \forall v_1, v_2 \in V$ and $\forall \alpha \in F$

(iii) $(\alpha_1 + \alpha_2) \cdot v = \alpha_1 \cdot v + \alpha_2 \cdot v \quad \forall v \in V$ and $\forall \alpha_1, \alpha_2 \in F$

(iv) $(\alpha\beta) \cdot v = \alpha(\beta v) = \beta(\alpha v) \quad \forall v \in V$ and $\forall \alpha, \beta \in F$

(v) $1 \cdot v = v \quad \forall v \in V$

(vi) $\alpha \cdot 0 = 0 \quad \forall \alpha \in F$

(vii) if $v \neq 0$, then $\alpha \cdot v = 0 \Rightarrow \alpha = 0$

(viii) $0 \cdot v = 0 \quad \forall v \in V$

(ix) if V is a vector space over a field F , then any linear combination of vectors lying in V (with scalars from F) would again lie in V

(10.5)

(100 H) let $V = \{x: \mathbb{R} \rightarrow \mathbb{C} \mid x \text{ is continuous and } x(t+T) = x(t) \forall t \in \mathbb{R}\}$
 → defining addition and scalar as usual
 $\hookrightarrow (x+y)(t) = x(t) + y(t)$
 $(\alpha x)(t) = \alpha(x(t))$ for $x, y \in V$ and $\alpha \in \mathbb{C}$
 To prove V is a vector space, proving —

(100 H) $(V, +)$ is an Abelian group

→ Closure under addition

let $x, y \in V$, Both x and y are continuous

and ~~continuous and periodic~~

$$x(t) = x(t+T), y(t) = y(t+T) \forall t$$

Then

$$(x+y)(t+T) = x(t+T) + y(t+T) = x(t) + y(t) = (x+y)(t)$$

so $x+y \in V$

→ Associativity

$$((x+y)+z)(t) = (x(t) + y(t)) + z(t) = x(t) + (y(t) + z(t)) = (x + (y+z))(t)$$

so $(x+y)+z = x+(y+z)$

→ Identity element

The zero (null) function $0: \mathbb{R} \rightarrow \mathbb{C}$ defined by $0(t) = 0$ is continuous and $0(t+T) = 0 = 0(t)$, Hence $0 \in V$

For any $x \in V$, $(x+0)(t) = x(t) + 0 = x(t)$ so

0 is an additive identity.

→ Commutativity

Positive commutativity of complex addition yields

$$(x+y)(t) = x(t) + y(t) = y(t) + x(t) = (y+x)(t)$$

Thus, $x+y = y+x$

→ Additive Inverse

$$\text{for } x \in V, (-x)(t) = -x(t), (-x)(t+T) = -x(t+T) = -x(t)$$

so $-x \in V$, $(x+(-x))(t) = x(t) - x(t) = 0$, so $x+(-x) = 0$

Thus, $(V, +)$ is an Abelian group

- ② Distributivity of scalar addition over vector
for $d_1, d_2 \in \mathbb{C}$ and $x \in V$

$$((d_1 + d_2)x)(t) = (d_1 + d_2)x(t) = d_1x(t) + d_2x(t) \\ = (d_1x + d_2x)(t)$$

$$\text{so } (d_1 + d_2)x = d_1x + d_2x$$

- ③ Distributivity of scalar over vector addition
 $d \in \mathbb{C}$ and $x, y \in V$

$$(d \cdot (x+y))(t) = d(x(t) + y(t)) = dx(t) + d(y(t)) \\ = (dx + dy)(t)$$

$$\text{so } d(x+y) = dx + dy$$

- ④ Compatibility of scalar multiplication (associativity)
for $\alpha, \beta \in \mathbb{C}$ and $x \in V$

$$((\alpha\beta)x)(t) = (\alpha\beta)x(t) = \alpha(\beta x(t)) = (\alpha(\beta x))(t)$$

$$\text{hence } (\alpha\beta)x = \alpha(\beta x)$$

- ⑤ Unit scalar

$$\text{for } x \in V, (1 \cdot x)(t) = 1x(t) = x(t) \text{ so } 1 \cdot x = x$$

- ⑥ $0 \cdot x = 0$ for all $x \in V$

$$(0 \cdot x)(t) = 0 \cdot x(t) = 0 \text{ so } 0 \cdot x \text{ is a zero function}$$

- ⑦ $0 \cdot x = 0 \neq x \in V$

$$(0 \cdot x)(t) = 0 \neq x(t) = 0$$

- ⑧ if $x \neq 0$ and $\lambda \cdot x = 0$ then $\lambda = 0$

Assume $x \in V$ is not the zero function, so $\exists t_0$ with $x(t_0) \neq 0$,
if $\lambda x = 0$ (the zero function), then

$$\lambda x(t_0) = 0, \text{ hence } \lambda x(t_0) = 0 \text{ and } x(t_0) \neq 0$$

$$\rightarrow \lambda = 0 \text{ (since } \mathbb{C} \text{ is a field)}$$

(ix) Closure under linear combinations.

Let $x_1, x_2, \dots, x_n \in V$ and $d_1, d_2, \dots, d_n \in \mathbb{C}$

$$y(t) = \sum_{k=1}^n d_k x_k(t)$$

Each x_k is continuous, so y is continuous.

and each x_k satisfies $x_k(t+T) = x_k(t)$

hence

$$y(t+T) = \sum_{k=1}^n d_k x_k(t+T) = \sum_{k=1}^n d_k x_k(t) = y(t)$$

Thus, any linear combination of vectors in V lies in V .

————— (2)

∴

Since all properties are satisfied,

all continuous time signals with time period

T form a vector space.

NOTE—

in (i), ~~remembering~~ ~~the~~ ~~and~~ wanting only some properties will fetch you only partial marks.

in (ii) for Abelian group, ~~the~~ The given proof should be present and wanting all 9 properties proofs only will fetch — (2) marks!